3D Vectors

Questions 1–28

Question 1. It is given that A, B and C have the coordinates $(-2, 3, 5)$, $(4, -1, -3)$ and $(-1, 0, -8)$ respectively. Find

a) $\overrightarrow{AB}$
b) $\overrightarrow{BC}$
c) $\overrightarrow{AC}$
d) $\overrightarrow{BA}$

Question 2. Find the magnitude of the following vectors:

a) $4\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$
b) $\begin{pmatrix} 6 \\ 7 \\ 1 \end{pmatrix}$
c) $5\mathbf{i} - 2\mathbf{j} + 12\mathbf{k}$
d) $\begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$

Question 3. Given each of the following position vectors, find the magnitude of the vector $\overrightarrow{AB}$.

a) $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}$

b) $\overrightarrow{OA} = \mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ and $\overrightarrow{OB} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$

c) $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 0 \\ -2 \\ 13 \end{pmatrix}$

Question 4. Find the unit vector in the direction of the following vectors:

a) $2\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$

b) $\begin{pmatrix} 6 \\ -1 \\ 4 \end{pmatrix}$

c) $8\mathbf{i} - 6\mathbf{j} + 10\mathbf{k}$

d) $\frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

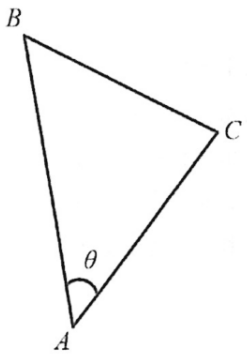
Question 5. Given each of the following position vectors, find the unit vector in the direction of the vector $\overrightarrow{AB}$:

a) $\overrightarrow{OA} = \begin{pmatrix} 4 \\ 8 \\ -2 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 6 \\ 0 \\ -2 \end{pmatrix}$

b) $\overrightarrow{OA} = 2\mathbf{i} - 10\mathbf{j} - \mathbf{k}$ and $\overrightarrow{OB} = 7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$

Question 6. It is given that $\overrightarrow{AB} = \begin{pmatrix} -1 \\ 2 \\ 8 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix}$ and the angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is θ. Show that $\cos\theta = \frac{10}{\sqrt{897}}$.

Question 7. The diagram shows the triangle ABC.



In each case, show that the value of the angle θ is the given value to the nearest degree:

a) $\theta = 18.3°$, $\overrightarrow{AB} = 2\mathbf{i} - 10\mathbf{j} + \mathbf{k}$ and $\overrightarrow{BC} = 7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$

b) $\theta = 135.5°$, $\overrightarrow{AB} = \begin{pmatrix} 1 \\ 3 \\ -7 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$

Question 8. Position vectors of the points A, B and C are given as $\overrightarrow{OA} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\overrightarrow{OB} = \mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $\overrightarrow{OC} = 8\mathbf{i} - 7\mathbf{j} - \mathbf{k}$. Given that $\overrightarrow{BC} = \overrightarrow{AD}$, find the position vector of the point D.

Question 9. Three position vectors are given to be

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ 6 \\ -1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 4 \\ 9 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 7 \\ 12 \\ p \end{pmatrix}$$

There is another point D and $\overrightarrow{AB} = \overrightarrow{BD}$.

a) Find the position vector of D.

b) Given $|\overrightarrow{AC}| = 9$, find the possible values of p.

Question 10. Show that the coordinates $(7, 4, 0)$, $(12, 6, -1)$ and $(-13, -4, 4)$ are collinear.

Question 11. Show that the coordinates $(6, 4, -3)$, $(8, 5, -3)$, $(12, 7, -15)$ are not collinear.

Question 12. Three points A, B and C lie on a line. It is known that $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 5 \\ 12 \\ 2 \end{pmatrix}$ and furthermore that $|\overrightarrow{AC}| = 3|\overrightarrow{BC}|$. Find the possible position vectors of the point C.

Question 13. The vector **a** is parallel to the given vector **p**. In each case, find the vector **a**:

a) $\mathbf{p} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $|\mathbf{a}| = \sqrt{84}$

b) $\mathbf{p} = -\mathbf{i} + 7\mathbf{j} + 2\mathbf{k}$ and $|\mathbf{a}| = \sqrt{6}$

c) $\mathbf{p} = \mathbf{i} - 8\mathbf{j} - \mathbf{k}$ and $|\mathbf{a}| = 3\sqrt{66}$

Question 14. Three position vectors are given as

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 24 \\ -3 \\ 7 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}$$

Find

a) the midpoint of AB, m_{AB}

b) the midpoint of BC, m_{BC}

c) the distance between m_{AB} and m_{BC}

d) a unit vector in the direction of the line passing through m_{AB} and m_{BC}

The point D lies a distance of 1 unit away from A parallel to the line passing through the midpoints.

e) Find the possible coordinates of D.

Question 15. For each of the following position vectors, state on which axes they lie:

a) $\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$

b) $-12\mathbf{i}$

Question 16. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ where

$$\mathbf{a} = \begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix}$$

A third point C has position vector $k(\mathbf{a} + k\mathbf{b})$ where $k \neq 0$ and it is known that C lies on the x-axis.

a) Find the value of k.

b) Write down the position vector of C.

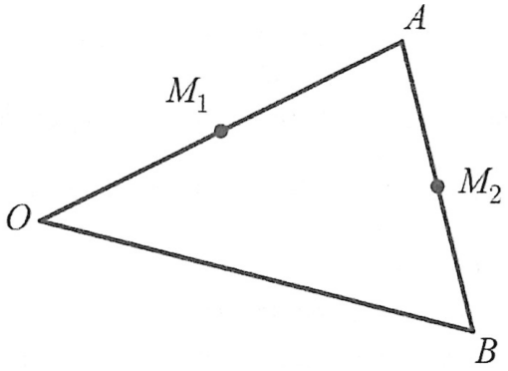
Question 17. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$. A point C lies on AB such that $AC : CB$ is $6 : 1$. Show that the position vector of C is $\frac{1}{7}(\mathbf{a} + 6\mathbf{b})$.

Question 18. Points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} = \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 14 \\ 7 \\ k \end{pmatrix}$$

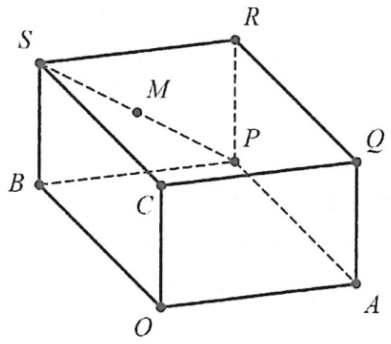
where k is a positive integer. It is known that $p\mathbf{a} + q\mathbf{b} = \mathbf{c}$ where p and q are positive integers. Find the values of p, q and k.

Question 19. The diagram shows the triangle OAB with vertices labelled. The midpoint of $\overrightarrow{OA}$ is M_1 and the midpoint of $\overrightarrow{AB}$ is M_2.



By considering O as the origin, so that $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$, show that the line joining M_1 and M_2 is parallel to $\mathbf{b}$ and is half the length of $\mathbf{b}$.

Question 20. The diagram shows the cuboid $OAPBCQRS$, with vertices labelled. The midpoint of $\overrightarrow{PS}$ is M.



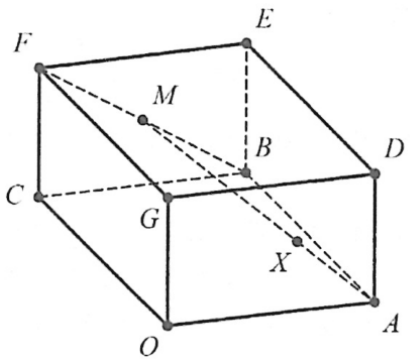
By considering O as the origin, so that $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$.

a) Show that $\overrightarrow{AM} = -\frac{1}{2}\mathbf{a} + \mathbf{b} + \frac{1}{2}\mathbf{c}$.

b) Find $\overrightarrow{PC}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

c) Given that the lines $\overrightarrow{AM}$ and $\overrightarrow{PC}$ intersect, find the position vector of the point of intersection in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

Question 20 (continued). Consider now, that the point X lies on AM such that $XM = 2AX$.



d) Find the position vector of X in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

Question 21. Three points ABC form an isosceles triangle such that $|\overrightarrow{AB}| = |\overrightarrow{BC}|$, and it is known that the position vectors of A and B are

$$\mathbf{a} = \begin{pmatrix} 1 \\ 8 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 5 \\ 7 \end{pmatrix}$$

respectively. The third point C has position vector $\mathbf{c} = \begin{pmatrix} 5 \\ 4 \\ k \end{pmatrix}$ where k is a positive constant to be found.

a) Find the possible values of k.
b) Find the exact area of ABC for both your values of k.

Question 22. Four points A, B, C and D have position vectors

$$\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}, \quad \mathbf{b} = 6\mathbf{i} + 5\mathbf{k}, \quad \mathbf{c} = 5\mathbf{i} + 5\mathbf{j} + 7\mathbf{k} \quad \text{and} \quad \mathbf{d} = \mathbf{i} + 6\mathbf{j} - \mathbf{k}$$

Show that $ABCD$ is a parallelogram.

Question 23. Three position vectors of points A, B and C are

$$\mathbf{a} = \begin{pmatrix} -1 \\ 5 \\ 8 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 4 \\ 13 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -2 \\ 5 \\ 19 \end{pmatrix}$$

There is another point D such that $ABCD$ forms a parallelogram and $\overrightarrow{AB} = \overrightarrow{DC}$.

a) Find the position vector of D.

b) Find the exact length of $\overrightarrow{AC}$.

Question 24. Three position vectors of points A, B and C are

$$\mathbf{a} = 2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}, \quad \mathbf{b} = 6\mathbf{i} + 12\mathbf{j} + \mathbf{k} \quad \text{and} \quad \mathbf{c} = -\mathbf{i} + \mathbf{j} + 4\mathbf{k}$$

There is another point D such that $ABCD$ forms a parallelogram and $\overrightarrow{AB} = \overrightarrow{DC}$.

a) Find the position vector of D.

b) Find the length of $\overrightarrow{AB}$.

c) Find the length of $\overrightarrow{AC}$.

The angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is θ.

d) Show that $\cos\theta = \frac{105}{\sqrt{12351}}$.

e) Find the area of the parallelogram $ABCD$.

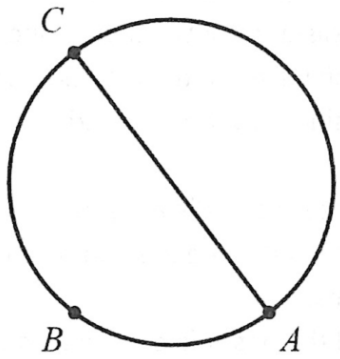
Question 25. Four points A, B, C and D have position vectors

$$\mathbf{a} = \mathbf{i} - 4\mathbf{k}, \quad \mathbf{b} = 3\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}, \quad \mathbf{c} = 8\mathbf{i} + 3\mathbf{j} - 10\mathbf{k} \quad \text{and} \quad \mathbf{d} = 6\mathbf{i} - 6\mathbf{k}$$

a) Show that $ABCD$ is a rhombus.

b) Find the area of the rhombus.

Question 26. The position vectors of points A and B are given by $\mathbf{a} = \mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$ and $\mathbf{b} = 7\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. The points A and B lie on a circle of radius $\sqrt{66}$. A further point $C(-1, -3, p)$ exists such that AC is the diameter of the circle.



a) Find the length of $\overrightarrow{BC}$.
b) Find the possible value(s) of p.

Question 27. Four position vectors of points A, B, C and D are

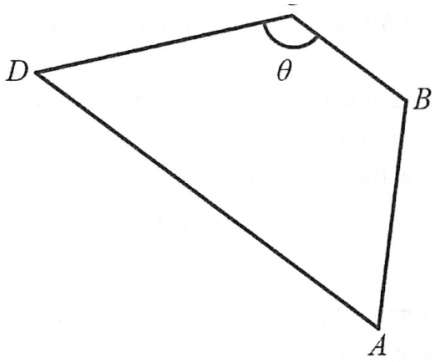
$$\mathbf{a} = -\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}, \quad \mathbf{b} = \mathbf{i} + 5\mathbf{j} + 12\mathbf{k}, \quad \mathbf{c} = -2\mathbf{i} + 10\mathbf{j} + 13\mathbf{k} \quad \text{and} \quad \mathbf{d} = -7\mathbf{i} + 14\mathbf{j} + 9\mathbf{k}$$

Show that $ABCD$ is a trapezium.

Question 28. Three position vectors of points A, B and C are

$$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 8 \\ -3 \\ 3 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 9 \\ -1 \\ 7 \end{pmatrix}$$

There is an additional point D such that $ABCD$ is a trapezium where $\overrightarrow{BC} = \frac{1}{3}\overrightarrow{AD}$.



a) Find the position vector of D.

The angle between BC and CD, θ, is given by $\cos\theta = -\frac{31}{7\sqrt{30}}$.

b) Show that the exact area of the trapezium $ABCD$ is $2\sqrt{509}$ units^2.